H

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 1

CANDIDATE NAME

BIOLOGY CLASS

REGISTRATION NUMBER

BIOLOGY

8875/02

Paper 2

30 August 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question.

<u>Section A</u>

Write your answers in this booklet using dark blue or black pen.

<u>Section B</u>

Write your answers on the separate Answer Paper using dark blue or black pen.

For Examiner's Use	
Paper 2	
Section A	
1	/ 10
2	/ 7
3	/ 12
4	/ 11
Section B	
5 / 6	/ 20
TOTAL	**/ 60**

This document consists of **12** printed pages.

[Turn over

Section A

Answer all the questions in this section.

1 Fig. 1.1 is a transmission electron micrograph of a mammalian cell.

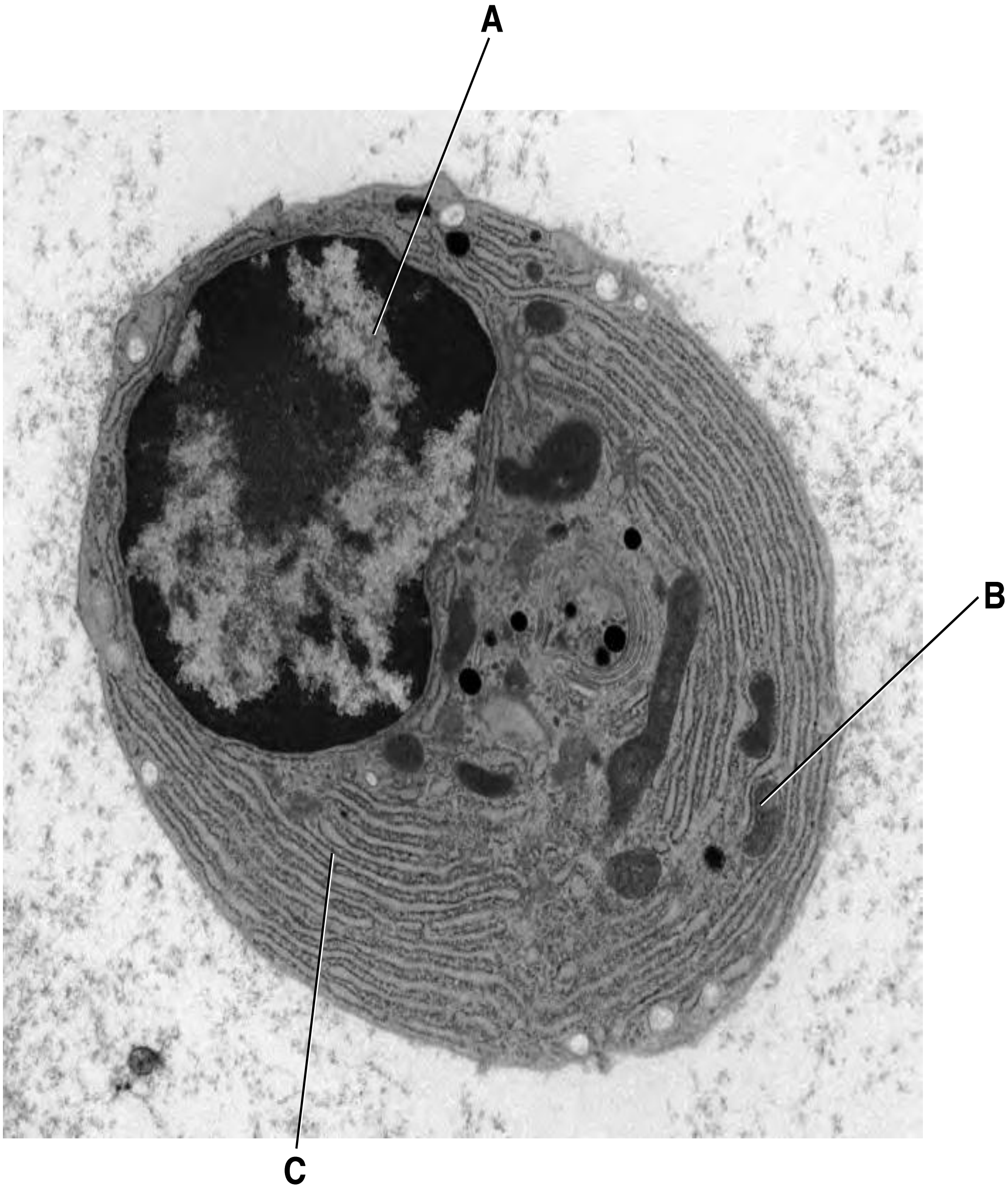

Fig. 1.1

(a) Complete Table 1.1 to:

- name in full, structures **A**, **B**, and **C.**
- outline how each structure functions to contribute to its specific role in the mammalian cell.

Table 1.1

structure	*name of structure*	*function of structure within mammalian cell*
A		
B		
C		

[3]

Table 1.2 below represents the ionic concentrations inside and outside a typical mammalian cell.

Table 1.2

	normal conditions	
	concentration (mM)	
ions	*intracellular*	*extracellular*
Na^+	5-15	145
H^+	7×10^{-5}	4×10^{-5}
Cl^-	5-15	110

(b) Explain how the intracellular concentration of Na^+ is maintained.

....

[1]

....

Cholera, a disease caused by a bacterial infection of *Vibrio cholerae,* results in severe diarrhoea leading to dehydration. A toxin released by the bacterium causes the release of chloride ions (Cl^-) from cells lining the small intestines into the lumen and inhibits the uptake of sodium ions (Na^+) by these cells. Under normal circumstances, the Cl^- intracellular concentration is maintained due to the natural electrical negativity of the cell.

(c) By comparing normal and diarrhoeal situations, explain how this disruption of cellular ion concentrations would result in extreme dehydration.

[4]

(d) One of the antibiotics used in the treatment of cholera is tetracycline. Tetracycline causes the death of the pathogen by binding to ribosomes and inhibiting protein synthesis.

(i) Suggest one way in which tetracycline acts at ribosomes to inhibit protein synthesis.

[1]

(ii) Tetracycline does not harm mammalian cells. Suggest an explanation for this.

[1]

[Total: 10]

2 **(a)** Pituitary dwarfism is a sex-linked inherited condition in humans in which affected individuals have very short limbs. The family tree, in Fig. 2.1, shows part of one affected family.

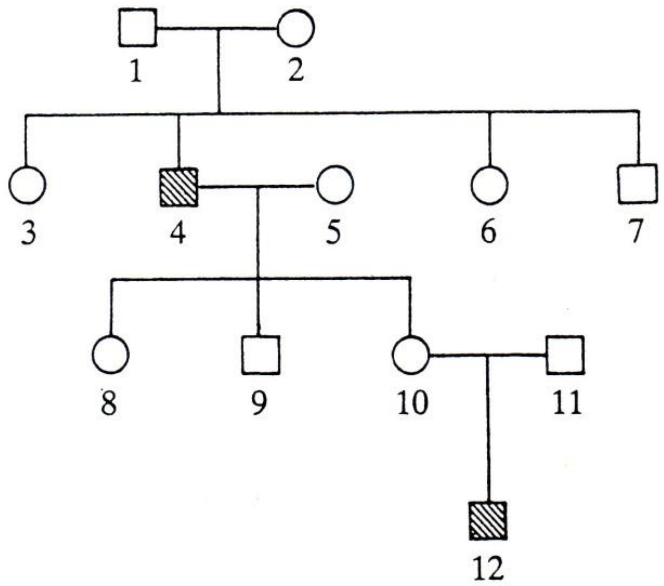

Fig. 2.1

(i) Explain what is meant by 'sex-linked inherited condition'.

[1]

(ii) With reference to Fig 2.1, explain the mode of inheritance using two pieces of evidences.

[4]

A series of breeding experiments were carried out with a certain species of mice. A cross between pure breeding males with red eyes and bent tails and females with black eyes and straight tails produced offspring all with red eyes and bent tails. The same results were obtained when the parents of the same pure –breeding phenotypes were used again but with the sexes reverse.

(b) Suggest a reason for carrying out the cross with the sexes reversed.

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.... [2]

[Total: 7]

3 Fig. 3.1 illustrates what happens on thylakoid membrane during daytime.

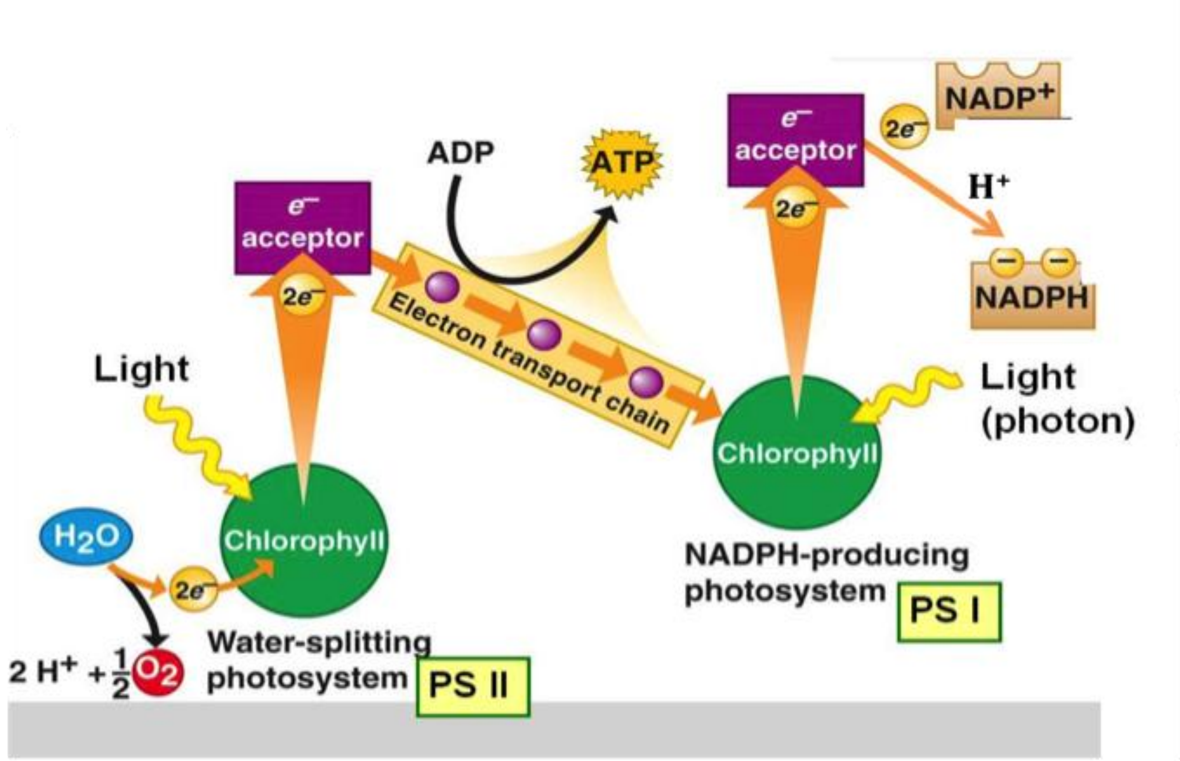

Fig. 3.1

(a)

(i) Describe how electron transport chain leads to ATP production.

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.... [3]

(ii) Compare the initial electron donor and final electron acceptor between oxidative phosphorylation and non-cyclic photophosphorylation.

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.... [2]

(b) A point mutation results in dysfunctional NADP$^+$ reductase. State and explain how the concentration of Ribulose 1,5 – Bisphosphate in the chloroplasts would be affected.

[3]

(c) Glyceraldehyde 3-phosphate exits the Calvin cycle for production of glucose, which will be used for cell wall formation. Explain how the structure of cellulose allows it to serve its function.

[4]

[Total: 12]

4 The corn borer, *Ostrinia nubilalis,* is an insect pest of maize. The larvae are caterpillars that eat the leaves of the maize plants. The adults can fly. Adult corn borers do not feed on maize plants.

Much of the maize that is grown in the USA has been genetically modified to produce *Bt* toxin, which is lethal to insects that feed on the leaves. However, many populations of the corn borer have now evolved resistance to the *Bt* toxin.

(a) Explain how this resistance could have evolved.

[3]

Fig. 4.1 shows the male and female flowers of maize.

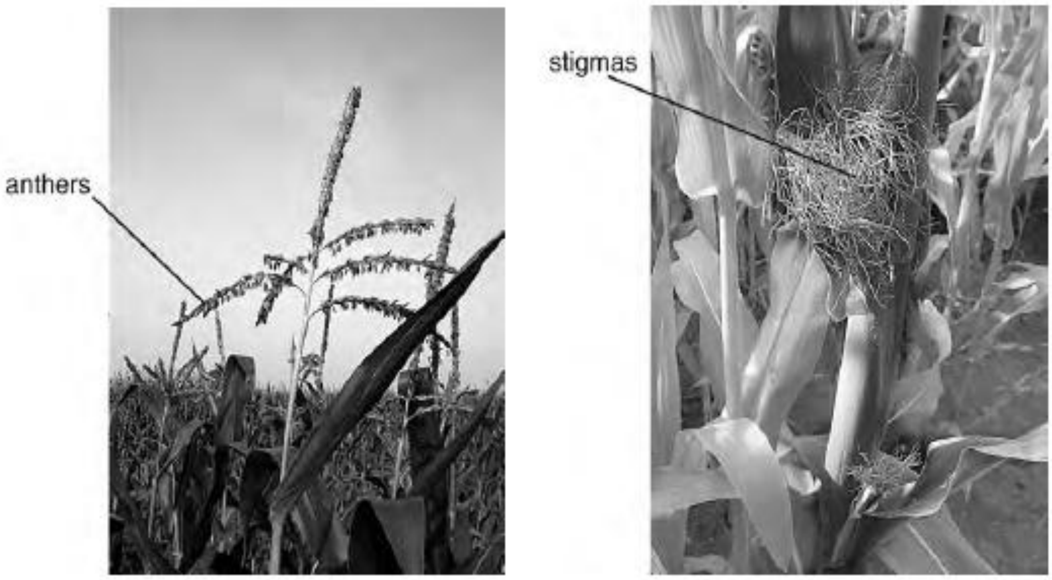

Fig. 4.1

Cultivated maize is related to a wild Mexican grass, teosinte. Teosinte looks very different from maize. In particular, teosinte has a very hard layer surrounding the fruit, making it impossible to use as an edible grain.

Archaeological excavations have found that maize in its edible form dates back at least 4000 years. It has been argued that the structural differences between teosinte and maize grains are so great that it is unlikely that maize could have been bred from teosinte so long ago.

Investigations have been carried out into the genetic differences between teosinte and maize. A gene on chromosome 4, *tga-1*, containing 1042 base pairs, was found to be responsible for all of the structural differences between teosinte and maize.

Fig. 4.2 shows the DNA base sequence of the only part of *tga-1* that always differs between teosinte and maize.

teosinte	GAT	TGG	GAT	CTC	AAG	GCG	GCG	GGC	GCG	TGG
maize	GAT	TGG	GAT	CTC	AAC	GCG	GCG	GGC	GCG	TGG

Fig. 4.2

(b) Outline the principles of electrophoresis as used in sequencing this DNA.

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.... [3]

(c) Suggest how the difference in the base sequence of the *tga-1* gene shown in Fig. 4.2 could cause large differences in phenotype between teosinite and maize.

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.... [2]

(d) Account for the role of **three** types of RNA in producing the protein expressed from *tga-1* gene.

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.... [3]

[Total: 11]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a), (b)** etc., as indicated in the question.

5 **(a)** With reference to specific genes, discuss how point mutations and chromosomal mutations can lead to cancer. [12]

(b) Describe homologous chromosomes and distinguish their behaviors between mitosis and the first division of meiosis. [8]

6 **(a)** Outline the procedure for cloning an eukaryotic gene in a bacterial pUC19 plasmid using *Escherichia coli* cells to produce eukaryotic proteins. Your answer should also include the method used to avoid intron-associated problem, the transformation of bacteria and the identification of the desired bacteria colonies. [12]

(b) Discuss the sources of variation important for natural selection to occur. [8]

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